

Assessment Tool-1:

Short Answer Test

2) Online Shopping System

- a) Apply Abstraction and Modularity to Divide the System into Components

Abstraction:

Abstraction hides unnecessary implementation details and exposes only essential features to users.

Example:

Customer → Browse products, add to cart, place orders.

Admin → Manage products, inventory and users.

Payment → Processes payment without exposing transaction details.

Modularity:

The system is used for the purpose to the sequence of the specific functionality.

The system is divided into independent modules, each responsible for a specific functionality.

Module	Responsibility
User Management	Registration, Login, Profile
Product Management	Add, Update, Delete Products
Shopping Cart	Manage Cart Items
Order Management	Create and Track Orders
Payment Module	Process Payment
Inventory Module	Update Product Stock
Notification Module	Send Email/SMS Notifications.

b) Apply SRP and OCP in Designing System Classes

Single Responsibility Principle [SRP]

Each class should have only one responsibility.

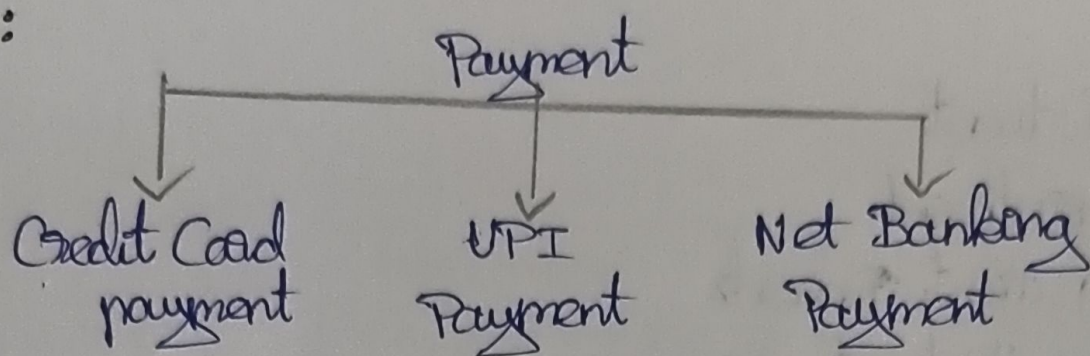
Class	Responsibility
Customer	Manage customer information
Product	Store product details
Cart	Manage shopping cart
Order	Handle order processing
Payment	Process payments
Notifications	Send notifications.

Benefits: Changes in one feature do not affect other classes.

Open-Closed Principle [OCP]

Classes should be open for extension but closed for modification.

Example:



When adding a new payment method, simply create a new subclass without modifying the existing Payment class.

5) How These Principles Improve Maintainability

- * Easier to understand because each module has a specific purpose.
- * Bugs are isolated within individual modules.
- * New features can be added without changing existing code.
- * Code becomes reusable and scalable.
- * Testing and debugging become simpler.
- * Future maintenance cost is reduced.

5) Hospital Management System

a) Apply UML Diagrams to Represent System Requirements.

UML diagrams help visualize the structure and behavior of the Hospital Management System.

Use Case Diagrams

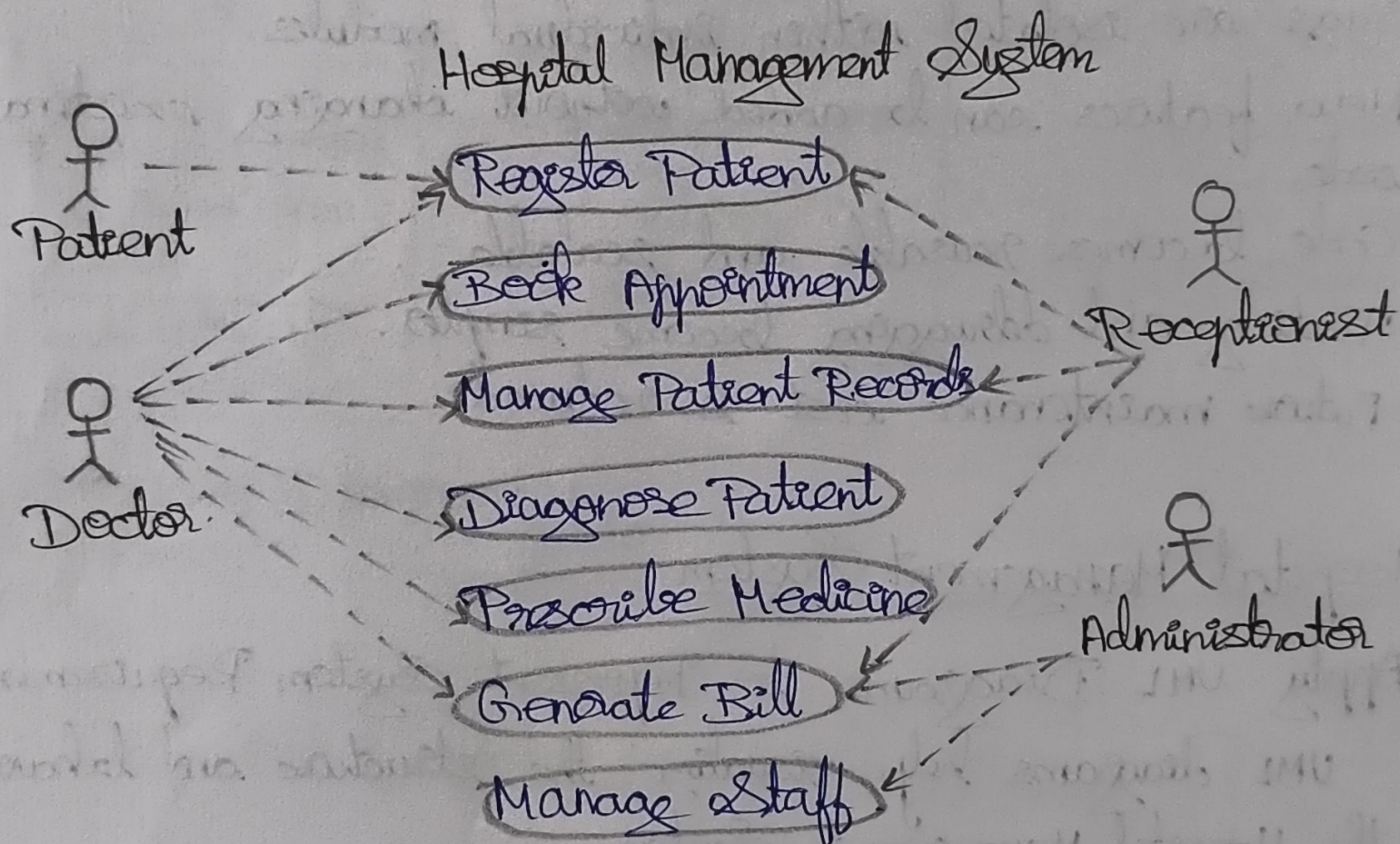
Actors

- * Patient
- * Doctor
- * Receptionist
- * Administrator

Use Cases:

- * Register Patient
- * Book Appointment
- * Manage Patient Records
- * Diagnose Patient
- * Prescribe Medicine
- * Generate Bill
- * Manage Staff

This diagram captures the functional requirements of the system.

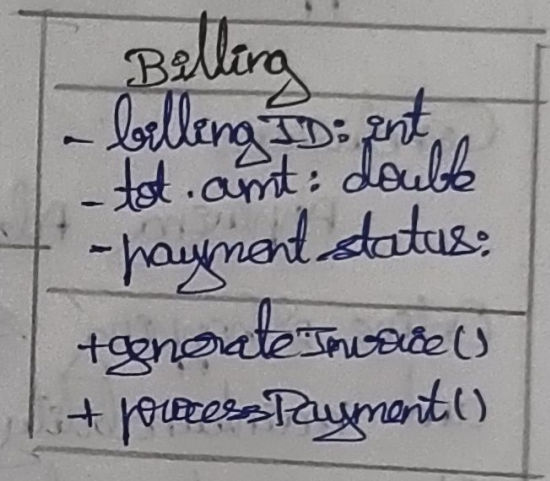
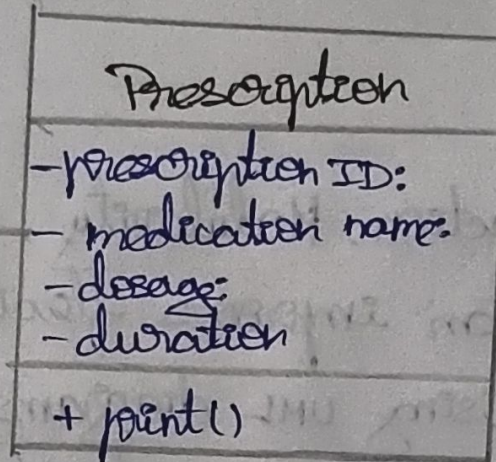
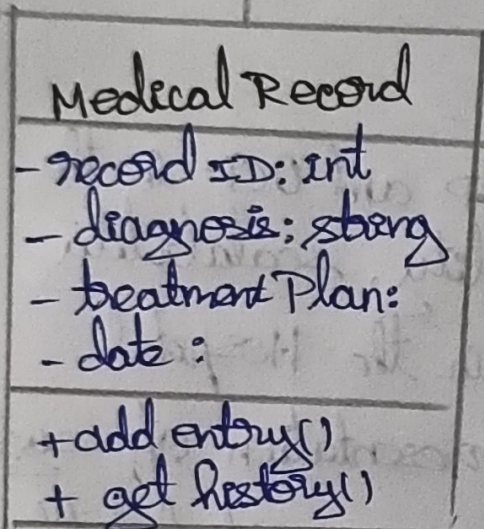
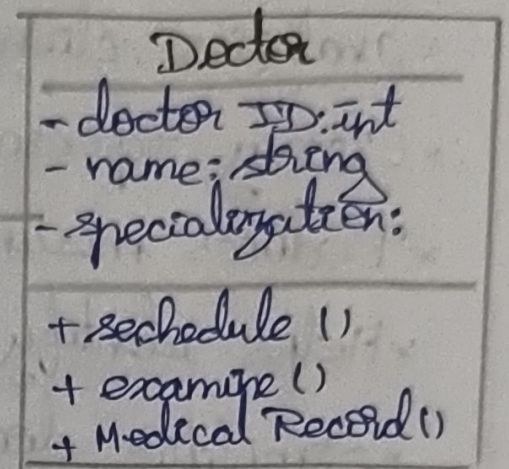
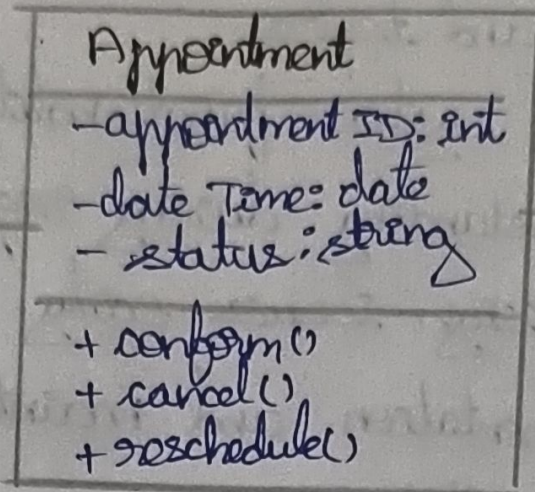
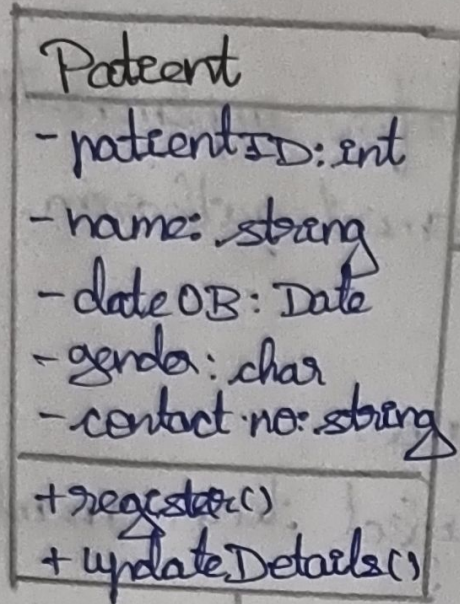


b) Demonstrate the Use of Class and Sequence Diagrams
Class Diagram

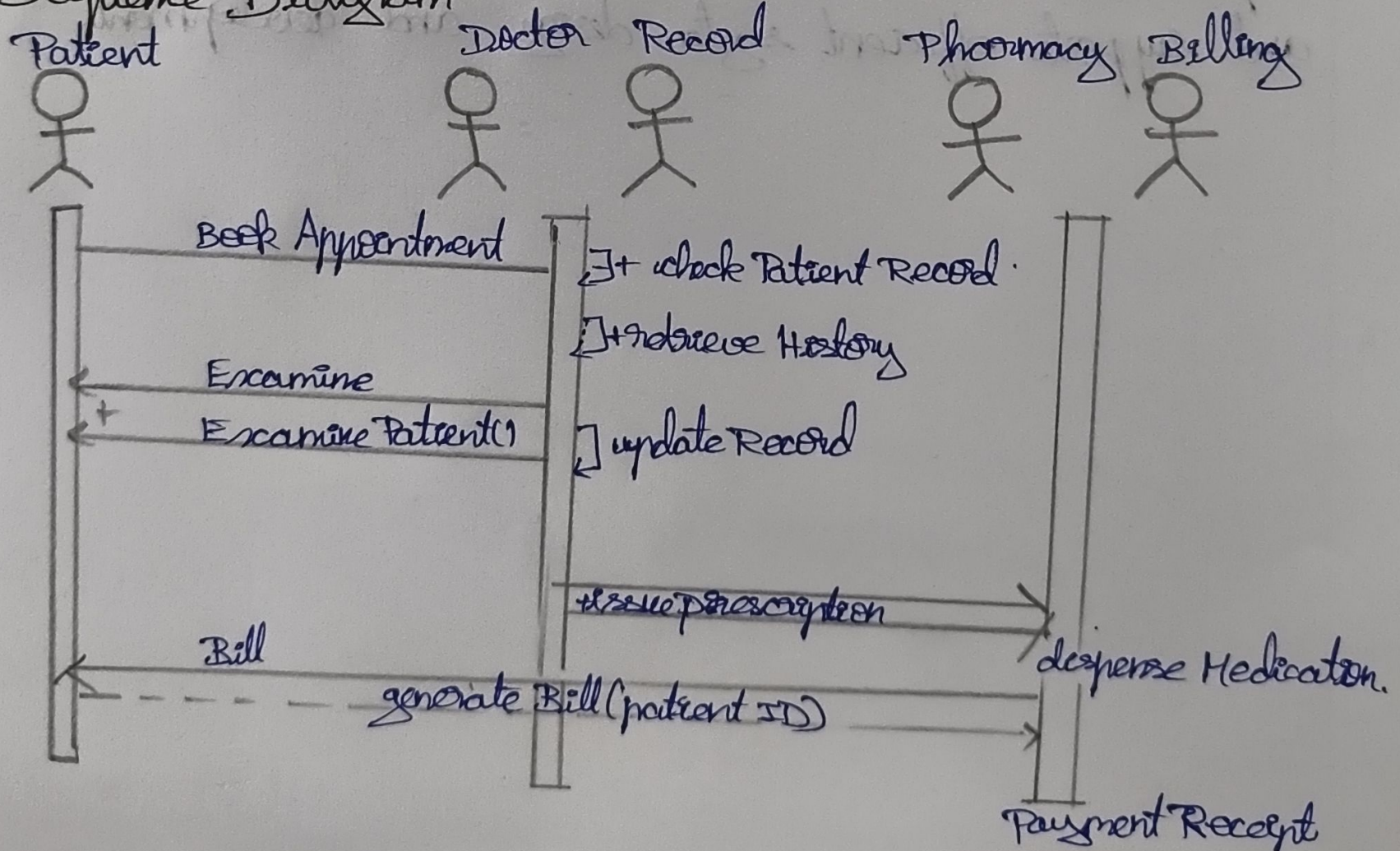
Classes.

- * Patient
- * Doctor
- * Appointment
- * Medical Record

Billing Prescription



Sequence Diagram



c) How UML improves communication among stakeholders

- * Provides a common visual language for developers, analysts, clients and testers.
- * Clearly represents system functionality before implementation
- * Reduces misunderstanding during requirement gathering.
- * Helps identify design issues early.
- * Simplifies documentation and maintenance.
- * Supports better collaboration among project team members.

Conclusion:

Applying Abstraction, Modularity, SRP and OCP in the Online Shopping system improves flexibility, scalability and maintainability. Using UML diagrams in the Hospital Management system provides a clear representation of system requirements, enhances communication among stakeholders and supports efficient system design and development.